

Institute of Actuaries of India

Subject CB1-Business Finance

March 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

<u>Solution 1:</u>	A	[2 Marks]
<u>Solution 2:</u>	D	[2 Marks]
<u>Solution 3:</u>	D	[2 Marks]
<u>Solution 4:</u>	D	[2 Marks]
<u>Solution 5:</u>	A	[2 Marks]
<u>Solution 6:</u>	A	[2 Marks]
<u>Solution 7:</u>	D	[2 Marks]
<u>Solution 8:</u>	C	[2 Marks]
<u>Solution 9:</u>	B	[2 Marks]
<u>Solution 10:</u>	D	[2 Marks]

Solution 11: The working capital cycle of the firm is the time from payment to suppliers for the materials to receiving cash from the sale of goods produced from the materials.

It is equal to inventory turnover period + trade receivables turnover period – trade payables turnover period, where:

- Inventory turnover period is the average time taken to sell an item of inventory after it has been purchased
- Trade receivable turnover period is the time taken to collect payment from a credit customer
- Trade payable turnover period the time that the company takes to pay for goods after their purchase

Negative working capital cycle may imply one or more of the following:

- The company is selling stock and getting paid for it more quickly than it is paying for the goods it buys on credit
- The business is running efficiently
- It could also mean that suppliers are being used as a source of finance (which could lead to strained relationships with suppliers)

[1 mark for definition of working capital cycle; 2 marks for definitions of its components; 2 marks for drawing inference around negative working cycle]

[5 Marks]

Solution 12: While B may be right that company is generating healthy “accounting profits”, truly, profit could look healthier than cash because:

- Full value of goods sold on credit is credited as revenue while preparing accounts, and will therefore contribute to accounting profit, even though cash may not be received in the same accounting period
 - This might be particularly true because the company is new and might be aggressive on credit sales to tap into the market
- As per accruals concept, inventories (the stock of materials, components, work-in-progress and finished goods) are treated as a cost only when they are used to produce the goods sold
- Hence, any build-up of inventories will reduce cash by more than it will reduce profit

- Further, cash balances are affected by items that do not affect the statement of profit or loss, e.g. a non-current asset may have been purchased by the company

Cause for concern

- A new business may show substantial profits (especially niche industries like chemicals, etc.) and yet experience large cash outflows as it invests in new equipment, builds up stock and sells on credit.
- Hence, B may be right that the discrepancy between the profit position and the cash position is not a cause for concern.
- However, despite high profit margins and efficient operations, business might fail if they run out of cash and/or exceed their overdraft limits.
- Hence, A and B must forecast and monitor their cash positions and should take action to remedy any liquidity problems.

[3.5 marks for explaining why profits could look healthier than cash; 1.5 marks to draw conclusions]

[5 Marks]

Solution 13: The preparation of insurance company accounts is complicated by two special features:

- The underlying contracts (liabilities) fall due outside the accounting period and are uncertain in size
- Premature transfer of 'profit' to shareholders may endanger the financial stability of the company and the ability to meet future liabilities

Additional items that appear in the statement of financial position of an insurance company are as follows:

Assets

- Assets held to cover insurance liabilities
- Assets representing free reserves
- Reinsurers' share of technical provisions
- Trade receivables arising out of direct insurance operations
- Trade receivables arising out of reinsurance operations
- Prepayments and accrued income

Liabilities

- Fund for future appropriations
- **Technical provisions:**
 - Long-Term Insurance business provisions, including the actuarially estimated value of the company's liabilities including bonuses already declared and after deducting the actuarial value of future premiums
 - General Insurance business provisions, including unexpired risk reserves and outstanding claims reserve

[2 marks for providing reasons of complexity; 2 marks for listing additional items on Assets side of Balance Sheet; 1 Mark for listing additional items on Liability side of Balance Sheet – Total 5 Marks]

[5 Marks]

Solution 14: Historical cost accounting tends to distort profits during times of inflation:

a) Valuation of inventories

- There is a time lag between the purchase of raw materials and its eventual sale of processed good.
- This means that companies are constantly understating costs of sales.
- If, for example, an item is purchased for INR 100 and sold for INR 150, the company will record a profit of INR 50.
- If, however, the cost of replacing that item of inventory has increased to INR 120 (due to inflation, let's say), then the 'real' profit on the transaction is only INR 30.
- Hence, this accounting practise tends to overstate profits.

b) Depreciation

- The depreciation charge is calculated using the historical cost of the assets.
- If inflation is positive, straight line depreciation over the useful life of a machine purchased, say, seven years ago will be less than on an identical machine purchased, say, two years ago.
- The difference will be particularly significant when inflation is high.
- Hence, historical cost accounting tends to overstate profits in times of positive inflation.

c) Interest payments

- A company may:
 - Receive interest on its investments, and
 - Pay interest on its loan capital.
- In times of inflation, part of the interest payment is really a compensation for the erosion of the capital.
- Hence, for a company that pays out more interest than it receives, profits will tend to be understated in times of high inflation.
- Similarly, for a company that pays out less interest than it receives, profits will tend to be overstated in times of high inflation.

d) Consistency over time

- Profits and asset values might be increasing in money terms.
- However, it would not be obvious how much was due to a real increase in the scale of a company's operations and how much was simply due to inflation.
- Comparison between years is therefore difficult when using historical cost accounts because inflation will affect different figures in the financial statements in a different way.
- This sort of criticism applies to many items where values are compared over time.

[0.5 mark for every valid sub-point; maximum 1.5 marks for any one heading i.e. full marks can be obtained if all four headings have been reasonably explained]

[5 Marks]

Solution 15: The scrip issue of MTG Ltd offers the investor a choice between Rs 5.75 cash and
 $2 \times 27/9 = 6$ of equity.

At face value, the scrip dividend appears to have more value, although there will be some dilution.

Transaction costs need to be considered while accepting the scrip dividend with a view to resale the same immediately.

The investor should also consider if there is any tax on the scrip dividend or cash dividend. The shareholder will have to consider whether the scrip issue will leave the portfolio unbalanced.

[2 marks for point number one and 1 mark for each of the subsequent points or any other valid points]

[5 Marks]

- Solution 16:**
- Opportunity cost is the cost of not taking up the next best investment choice when making a financial decision. It is the return forgone on alternate investments of equivalent risk.
 - If a company has a number of mutually exclusive projects then the opportunity cost of accepting one project is that it will be impossible to accept any of the others.
 - Companies might also face capital rationing. Accepting a positive NPV project might involve using funds that could be applied more profitably elsewhere.
 - Considering opportunity costs make decision making little complex because of the need to determine the NPV of competing projects. However, it may be difficult to spot opportunities that have been missed or overlooked and so there may be very little risk of criticism arising from a missed opportunity.
 - A company can't use the same opportunity cost for all the projects. The risk differs across projects especially if they are from diverse fields, therefore the opportunity cost will be different across projects.

*[1 mark for every valid point maximum 1 marks for any example explained.
 full marks can be obtained if all five headings have been reasonably explained]*

[5 Marks]

- Solution 17:**
- Maximisation of shareholder wealth is a simple and effective objective for evaluating managerial decisions.
 - The concept of increases shareholder wealth is clearly in line with shareholders objectives, and even allows for the interaction between risk and return. This is theoretically correct.
 - In practice, it may be difficult to ensure the directors acting in this way.
 - There are many ways in which maximising shareholder wealth could be against the directors' best interests. For example, shareholders can diversify risks in ways that directors cannot and so directors may be more risk averse in decision making than shareholders would prefer. Many directors have sufficient integrity to overcome any such pressures, but it will be difficult to distinguish them from those who have not.

- Wealth can be created for shareholders at the expense of other stakeholders if the company subjects other stakeholders to greater risks than what they envisaged or imposes damages on society, the costs of which can't be traced back to the company.

[1 mark for every valid point maximum 1 marks for any example explained. full marks can be obtained if all five headings have been reasonably explained]

[5 Marks]

Solution 18:

- Real Estate companies raise a high proportion of their long term financing from debt rather than from equity.
- The main reason for the same is that a real estate company can offer a lender security against its assets.
- That means that a lender would be less concerned about the risk of default because the collateral should ensure that the capital will be repaid.
- A real estate company is also likely to raise a steady stream of income from tenants. That means that it can take on higher borrowing because it knows that it will be able to service them from rental income.
- The investment in property is likely to yield a steady capital gain over the long term, with a relatively low risk. The returns are unlikely to be high and raising funds from borrowing will be cheaper than using equity.

[1 mark for every valid point maximum 1.5 marks for any example explained. full marks can be obtained if all four headings have been reasonably explained]

[5 Marks]

Solution 19:

i)

- The company might use a standard discounting rate/ hurdle rate for all capital projects.
- This might consider the cost of the company's capital.
- It would also take account of the systematic risks associated with investing in the company as identified by the company's beta.
- Essentially, the discount rate should reflect the risks associated with the project and also the need to generate a real rate of return from the capital that has been invested.

[1 mark for every valid point. full marks can be obtained if all headings have been reasonably explained]

[3]

ii)

- IRR is the discount rate or the interest needed for the project to break even given the initial investment.
- If market conditions change over the years, this project can have multiple IRRs. In other words, long projects with fluctuating cash flows and additional investments of capital may have multiple distinct IRR values.
- The advantage to using the NPV method over IRR using the example above is that NPV can handle multiple discount rates without any problems.

Each year's cash flow can be discounted separately from the others making NPV the better method.

[1 mark for every valid point. full marks can be obtained if all headings have been reasonably explained]

[3]

iii)

- In theory, debt is cheaper than equity because lenders have lower risks compared to equity provider and so borrowing will reduce WACC because of the higher proportion of debt.
- In practice, this might be slightly more complicated because borrowing will affect the gearing levels and that will affect the risk characteristics of existing equity.
- Borrowing will increase the risk of holding shares and that could increase the cost of equity. At higher levels of gearing the risk attached to debt might also increase and that could increase the cost of borrowing too.
- These additional costs will offset the savings from using the cheaper source of finance.
- Borrowing also carries a tax advantage because interest can be offset against profit whereas dividends cannot.
- The Modigliani and Miller argument suggests that the reduction in WACC due to borrowing will be exactly offset by overall increases in the cost of equity.
- This means that there is no particular cost advantage in using debt or equity. However this argument ignores taxation.

[1 mark for every valid point. full marks can be obtained if all headings have been reasonably explained]

[7]

iv)

- The methodology used to appraise investment projects require many highly subjective decisions to be made and evaluated about the financial and non-financial parameters that are to be used in the model.
- In practice, that often means that any particular project can be justified by adjusting estimates about the projected cash flows and the appropriate discount rate.
- Companies may have formal investment appraisal systems but their implementation may be influenced by the political, social and regulatory environment.
- The quantification of such non-financial factors may be a challenge. Sometime decisions are taken basis long term strategic considerations.
- Apart from forecasting and capital considerations, arguments can be put forward to use more basic techniques such as payback period for certain projects or to dispense with formal appraisal altogether because the project is required for safety or for strategic purposes.
- Some projects may not be profitable independently, but may be undertaken because they are considered to benefit the company as a whole.

- Sometimes an investment project might be undertaken to achieve synergy with other projects undertaken by the company.
[1 mark for every valid point. full marks can be obtained if all headings have been reasonably explained]

[7]

[20 Marks]

Solution 20: Non-Current Assets Working

For **factory**, straight line method of depreciation is used.

Hence, annual depreciation for factory =
(Book Value as at 31st March 2019 – Estimated Residual Value) / (Estimated Useful Lifetime as at 31st March 2019)

$$\text{i.e. } (1,40,000 - 12,500) / (25-9+1) = 7,500$$

Book Value of factory as at 31st March 2020 = Book Value of factory as at 31st March 2019 less depreciation for the year

$$\begin{aligned} &= 1,40,000 - 7,500 \\ &= 1,32,500 \end{aligned}$$

[2 marks]

For **machinery**, reducing balance method of depreciation is used.

Hence, following equation holds for annual rate of depreciation (R):

Book Value as at 31st March 2019 * [(1 – R) ^ (Estimated Useful Lifetime as at 31st March 2019)] = Estimated Residual Value

$$\text{i.e. } 40,000 * [(1 - R) ^ (10-2+1)] = 5,400$$

$$\text{i.e. } R = 1 - (5,400 / 40,000) ^ (1/9)$$

$$\text{i.e. } R = 20\% \text{ (rounded to 2 decimal places)}$$

Book Value of machinery as at 31st Mar 2020 = Book Value of machinery as at 31st Mar 2019 less depreciation for the year

$$\begin{aligned} &= 40,000 - 20\% * 40,000 \\ &= 32,000 \end{aligned}$$

[3 marks]

Current Assets Working

Inventory Turnover Ratio = (Inventories/Revenue) * 365

$$\text{i.e. } 14 = (\text{Inventories} / 3,00,000) * 365$$

$$\text{i.e. } \text{Inventories} = 11,500 \text{ (rounded to nearest 100)}$$

[1 mark]

Current Ratio (CR) = Current Assets (CA) / Current Liabilities (CL)

Quick Ratio (QR) = (CA – Inventories) / (CL)

$$\text{i.e. } CR/QR = CA / (CA - \text{Inventories})$$

$$CR * (CA - \text{Inventories}) = CA * QR$$

$$CR * CA - CR * Inventories = CA * QR$$

$$CA * (CR - QR) = CR * Inventories$$

$$CA = (CR * Inventories) / (CR - QR)$$

$$CA = (1.8 * 11,500) / (1.8 - 1.3)$$

Hence, Current Assets (CA) = 41,400 (rounded to nearest 100)

[2 marks]

$$\text{Receivable Turnover Ratio} = (\text{Trade Receivable} / \text{Revenue}) * 365$$

$$\text{i.e. } 28 = (\text{Trade Receivable} / 3,00,000) * 365$$

$$\text{i.e. Trade Receivable} = 23,000 \text{ (rounded to nearest 100)}$$

[1 mark]

$$\text{Total Current Assets} = \text{Inventories} + \text{Trade Receivables} + \text{Cash/Bank}$$

$$\text{i.e. Cash/Bank} = 41,400 - 11,500 - 23,000$$

$$\text{Hence, Cash/Bank} = 6,900$$

[0.5 marks]

Liabilities Working (Other than Share Capital)

$$\text{Current Ratio (CR)} = \text{Current Assets (CA)} / \text{Current Liabilities (CL)}$$

$$\text{i.e. CL} = 41,400 / 1.8$$

$$\text{Hence, Current Liabilities} = 23,000 \text{ (rounded to nearest 100)}$$

[1 mark]

$$\text{Asset Utilization Ratio (AUR)} = \text{Revenue} / (\text{Share Capital} + \text{Reserves} + \text{Long Term Debt})$$

$$\text{i.e., Share Capital} + \text{Reserves} + \text{Long Term Debt} = \text{Revenue} / \text{AUR}$$

$$\text{Hence, Share Capital} + \text{Reserves} + \text{Long Term Debt} = 3,00,000 / 1.64 = 1,82,900 \text{ (rounded to nearest 100)}$$

[1 mark]

$$\text{Asset Gearing Ratio (AGR)} = \text{Long Term Debt} / (\text{Share Capital} + \text{Reserves} + \text{Long Term Debt})$$

$$\text{i.e. Long Term Debt} = \text{AGR} * (\text{Share Capital} + \text{Reserves} + \text{Long Term Debt})$$

$$\text{i.e. Long Term Debt} = 5\% * 1,82,900 = 9,100 \text{ (rounded to nearest 100)}$$

[1 mark]

Share Capital Working

$$\text{Profit Margin} = \text{Profit Before Interest and Tax (PBIT)} / \text{Revenue}$$

$$\text{i.e. PBIT} = \text{Revenue} * \text{Profit Margin}$$

$$\text{Hence, PBIT} = 3,00,000 * 5\% = 15,000$$

[1 mark]

$$\text{Interest Paid on Debt} = \text{Interest Rate} * \text{Long Term Debt}$$

$$\text{i.e. Interest Paid} = 10\% * 9,100 = 900 \text{ (rounded to nearest 100)}$$

[0.5 marks]

$$\text{Profit Before Tax (PBT)} = \text{PBIT} - \text{Interest Paid}$$

$$\text{Hence, PBT} = 15,000 - 900 = 14,100$$

[0.5 marks]

$$\text{Profit After Tax (PAT)} = \text{PBT} * (1 - \text{Tax Rate})$$

$$\text{Hence, PAT} = 14,100 * (1 - 30\%) = 9,900 \text{ (rounded to nearest 100)}$$

[0.5 marks]

$$\text{Earnings Per Share (EPS)} = \text{Profit After Tax (PAT)} / \text{No. of Ordinary Shares}$$

i.e. No. of Ordinary Shares = PAT / EPS = 9,900 / 0.15 = 66,000

[1 mark]

Nominal Value Per Share = 0.25

Hence, Share capital = No. of Ordinary Shares * Nominal Value Per Share =
66,000 * 0.25 = 16,500

[0.5 mark]

Share Capital + Reserves + Long Term Debt = 1,82,900

i.e. Reserves = 1,82,900 – Long Term Debt – Share Capital

i.e. Reserves = 1,82,900 – 9,100 – 16,500 = 1,57,300

[0.5 marks]

Final Balance Sheet post consolidating all numbers generated above:

ASSETS:

a) Non-Current Assets

Factory	1,32,500
Machinery	32,000

b) Current Assets

Inventories	11,500
Trade Receivables	23,000
Cash/Bank	6,900
Total Assets	2,05,900

EQUITY AND LIABILITIES:

a) Total Equity

Share Capital (25p)	16,500
Other Reserves/Retained Earnings	1,57,300

b) Current Liabilities

23,000

c) Non-Current Liabilities

10% Loan	9,100
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Total Liabilities

2,05,900

[3 marks]

[20 Marks]
